MICROBIOLOGY

E. coli, Klebsiella, Proteus & Enterobacter spp.

SGL 1 · Dr Samira Fattah
GUE BLOCK

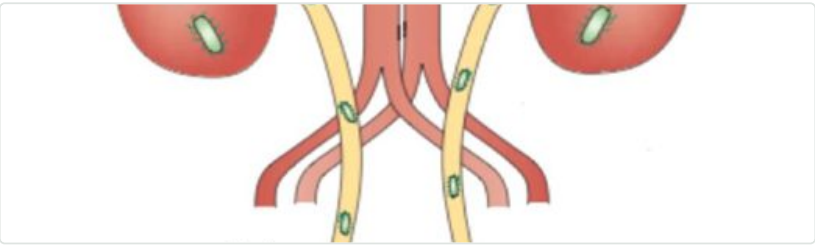
Microbiology GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

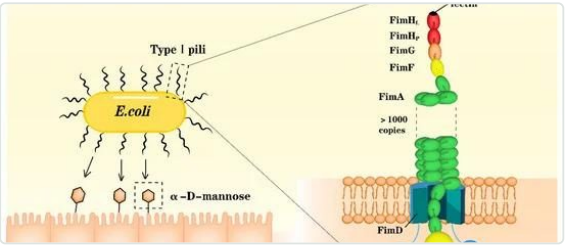
1. Which of these four organisms is the only one that is urease-NEGATIVE, and which two are strongly urease-POSITIVE?
2. Why does **Proteus** urease activity predispose to staghorn calculi, and which antibiotic class is unsuitable for it?
3. Which organism is notorious for AmpC β-lactamase overproduction, and which cephalosporin should generally be avoided for invasive infection with it?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

Shared Enterobacterales features — all 4 organisms: order **Enterobacterales**, Gram-negative non-spore-forming rods, facultative anaerobes fermenting glucose, usually **oxidase-negative**, reduce nitrate. Gut carriage is the **reservoir** for urinary infection — organisms reach the periurethral area, **ascend** the urethra, colonize the bladder (adhesion resists urine flow), and can further ascend to kidneys/bloodstream.



Ascending UTI pathway — periurethral colonization → bladder → kidney/bloodstream



Fimbrial (Type 1 pili) adhesion to colonic/uroepithelial cells

1 Clinical syndromes & complicating factors

Syndrome	Typical findings
Cystitis	Dysuria, **frequency**, urgency, suprapubic discomfort
Pyelonephritis	**Fever**, flank pain, costovertebral angle tenderness
Acute bacterial prostatitis	Fever, pelvic/perineal pain, urinary symptoms
Urosepsis	Urinary infection with **life-threatening organ dysfunction**

Complicating factors: obstruction/stones/incomplete emptying · urinary catheters/instrumentation · pregnancy, impaired immunity or renal impairment · recent antibiotics or a prior resistant isolate.

2 Escherichia coli

Morphology	Gram-negative rod; **usually motile** and lactose fermenting; **indole positive**, urease negative. Uropathogenic strains = **UPEC**.
Virulence	**Fimbriae** (bladder attachment/invasion) · capsule & surface structures (resist host defenses) · hemolysin/toxins (injure host cells) · **LPS** (triggers inflammation)
Disease	Commonest cause of acute **cystitis**; ascending infection → pyelonephritis; prostatitis (fever + pelvic pain); can progress to bacteremia/sepsis
Lab ID	MacConkey → **pink** lactose-fermenting colonies; indole usually +, citrate usually −. Colony appearance alone **cannot** establish the UPEC pathotype — AST needed when culture is clinically indicated.
Treatment	**Nitrofurantoin** for bladder infection only; **fosfomycin** an option for selected cystitis; ESBL-producing → TMP-SMX or a fluoroquinolone may suit complicated UTI



E. coli — Gram-negative bacilli/rods on Gram stain



Klebsiella — positive string test (viscous string lifted with a loop)

3 Klebsiella (K. pneumoniae)

Morphology	**Encapsulated** Gram-negative rod; classically **nonmotile** and lactose fermenting; colonies often **mucoid**; some isolates **string-test positive**
Virulence	**Polysaccharide capsule** (resists phagocytosis/host killing) · fimbriae (epithelial/device attachment) · biofilm (persistence on urinary devices)
Disease	Cystitis, pyelonephritis, urinary-source bacteremia; UTI after **catheterization**/instrumentation; colonizes the intestine, opportunistic infections
Resistance	Expected resistance to **ampicillin**; hospital exposure + recent antibiotics raise resistance concerns; **ESBLs and carbapenemases** may restrict treatment

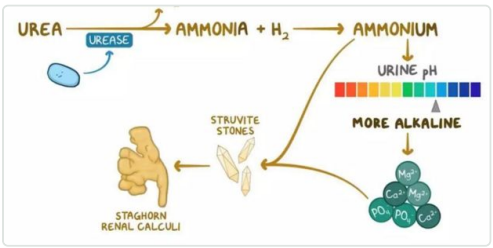
4 Proteus spp.

Morphology	**P. mirabilis** = principal urinary species (P. vulgaris also opportunistic); highly motile, typically **non-lactose fermenting**; **swarming** → concentric growth on blood agar
Key feature	Strong **urease** activity — urea + water → ammonia + CO_2 → urinary pH rises → alkaline conditions precipitate **struvite & carbonate apatite** → stones/catheter encrustation
Complications	Stones **protect bacteria** → recurrence; struvite stones may form a **staghorn calculus**; obstruction impairs drainage & damages the kidney; catheter biofilm + urease → crystalline encrustation & retention
Management	Choose agent by AST + site; **nitrofurantoin is unsuitable** (lecture poses this as a discussion question — ask your lecturer for the mechanism); persistent/recurrent infection raises concern for stones; restore drainage if obstructed

Long-term catheter bacteriuria **alone is not an indication for antibiotics** — catheter problems and infection need clinical assessment first.



Proteus — swarming, concentric growth on blood agar



Urease pathway → alkaline urine → struvite precipitation



Staghorn calculus filling the renal pelvis/ calyces

5 Enterobacter spp. (E. cloacae complex)

Identity	**E. cloacae complex** = several related species (E. cloacae, E. hormaechei, E. kobei, E. roggenkampii, E. asburiae) — routine testing could not tell them apart; newer methods (multiplex PCR, MALDI-TOF) are closing the gap
Morphology	Usually **motile**, glucose fermenting, **often lactose-fermenting**; colonizes the gut, persists in healthcare environments
Disease	Catheter-associated UTI, bacteremia, other opportunistic infections
Resistance	**AmpC** overproduction risk — AmpCs hydrolyze 3rd-gen cephalosporins and piperacillin/tazobactam. **Ceftriaxone generally avoided** for invasive infection; **cefepime** (4th-gen) may be appropriate if active & correctly dosed; carbapenem may be needed for severe disease

6 Discriminator grid — comparative lab identification

Feature	E. coli	K. pneumoniae	P. mirabilis	E. cloacae complex
Lactose	+	+	−	Usually +
Motility	+	−	+	+
Indole	+	−	−	Usually −
Urease	−	+	**Strong +**	Variable
Citrate	−	+	+	+
Useful clue	Commonest UTI cause	Mucoid colonies	**Swarming**	**AmpC risk**

7 Treatment & special situations

Situation	Key point
Nitrofurantoin	Bladder infection only; **unsuitable for Proteus**
Fosfomycin	Option for selected E. coli cystitis
ESBL-producing	TMP-SMX or a fluoroquinolone may be suitable in complicated UTI
Inducible AmpC	Resistance may emerge with some cephalosporins
Asymptomatic bacteriuria	Most nonpregnant adults (incl. older adults, diabetics, catheter users) **do not benefit** from treatment. Key exceptions: **pregnancy** and urological procedures with mucosal trauma
Pregnancy	Screen & treat ASB; choose antibiotics by gestation/susceptibility/safety; pyelonephritis in pregnancy needs prompt assessment

Empirical selection: identify the site (bladder/kidney/prostate/bloodstream) → assess severity (sepsis, obstruction) → review earlier cultures/recent antibiotics/local antibiogram → reassess & narrow after culture results.

P Must-remember pearls

- **Urease-positive + swarming** = Proteus → alkaline urine → struvite/staghorn calculi; nitrofurantoin does not cover it.
- **Mucoid colonies + positive string test** = Klebsiella; **expected ampicillin resistance**.
- **AmpC risk** is the Enterobacter discriminator — avoid ceftriaxone for invasive infection; cefepime or a carbapenem instead.
- Colony appearance on MacConkey can **never** establish the UPEC pathotype of an E. coli isolate — that needs virulence/genotype testing.
- Long-term catheter bacteriuria alone is **not** an indication for antibiotics — treat the patient, not the culture.

Q Test yourself

1. A 68-year-old man with a long-term catheter develops fever, flank pain and poorly draining urine. Urine pH is 8.5 and culture grows a swarming, urease-positive, indole-negative organism. What is the most likely complication to assess for?

A. Renal abscess from E. coli
B. Struvite/staghorn calculus
C. ESBL bacteremia from Klebsiella
D. AmpC-mediated treatment failure
E. Asymptomatic bacteriuria requiring no action

2. A 74-year-old woman with diabetes has a routine urine culture growing K. pneumoniae at 10^5 CFU/mL with pyuria, but no dysuria, fever or flank pain. What is the most appropriate next step?

A. Start nitrofurantoin empirically
B. Start an ESBL-active agent pending AST
C. No antibiotics — this is asymptomatic bacteriuria
D. Treat only if citrate is positive
E. Treat because diabetes is itself an indication

3. An ICU patient on a cephalosporin for an invasive Enterobacter cloacae complex infection initially improves, then deteriorates with recurrent bacteremia on the same drug. What best explains this?

A. The organism acquired an ESBL mid-treatment
B. AmpC overproduction induced by the cephalosporin
C. The organism was never Gram-negative
D. Nitrofurantoin resistance emerging
E. A new Proteus superinfection

4. A urine culture grows a lactose-fermenting, indole-negative, citrate-positive, nonmotile Gram-negative rod with mucoid colonies. Which organism is this, and what resistance should be assumed?

A. E. coli; expect fluoroquinolone resistance
B. Klebsiella pneumoniae; expect ampicillin resistance
C. Proteus mirabilis; expect nitrofurantoin resistance
D. Enterobacter; expect carbapenem resistance
E. Not identifiable without a string test

5. A pregnant woman at 28 weeks has a routine urine culture showing 10^5 CFU/mL E. coli with no urinary symptoms. What is the correct management?

A. No treatment — asymptomatic bacteriuria is not treated in pregnancy
B. Treat — pregnancy is a key exception where asymptomatic bacteriuria is treated
C. Repeat culture in 4 weeks only
D. Treat only if she develops fever
E. Treat only after delivery

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — Proteus urease hydrolyzes urea → ammonia → alkaline urine (pH 8.5) → struvite/carbonate apatite precipitation → staghorn calculus; swarming + urease-positive + indole-negative points to P. mirabilis.

2 — Most nonpregnant adults, including diabetics and catheter users, do not benefit from antibiotics for asymptomatic bacteriuria — pregnancy and urological procedures with mucosal trauma are the key exceptions, neither of which applies here.

3 — E. cloacae complex carries a meaningful risk of AmpC overproduction, which can be induced during cephalosporin therapy and cause clinical failure despite initial improvement — ceftriaxone is generally avoided for invasive infection for this reason.

4 — Lactose+, indole−, citrate+, nonmotile, mucoid = Klebsiella pneumoniae, which carries expected resistance to ampicillin.

5 — Pregnancy is one of the key exceptions where asymptomatic bacteriuria is actively screened for and treated, unlike in most nonpregnant adults.

SPACED REVIEW

☐ Day 1	☐ Day 3	☐ Day 7	☐ Day 21	☐ Pre-exam